

Healthcare Data Analysis - Exploratory Data Analysis (EDA)

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This project performs exploratory data analysis (EDA) on a healthcare dataset using Python and Jupyter Notebook.

The dataset includes information such as:

- Age
- Gender
- Diagnosis
- Date of Admission & Discharge
- Treatment Type
- Blood Pressure

Project Structure

- EDA_Healthcare_Report.pdf - PDF summary of all visual insights
- healthcare_dataset.csv - (Your dataset)
- eda_notebook.ipynb - Jupyter Notebook with code and plots
- README.pdf - Project overview and usage

Key Visualizations

1. Age Distribution: Understand the age profile of patients.
2. Gender Count: See the male-to-female ratio in the dataset.
3. Treatment Type Distribution: Explore the frequency of different treatments.

4. Diagnosis Frequency: Identify common and rare diagnoses.

5. Blood Pressure Analysis: Break down systolic and diastolic BP, and how it varies with age and diagnosis.

Technologies Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook
- FPDF (for PDF generation)

Getting Started

1. Clone the repo or open the notebook in Jupyter.
2. Make sure your dataset (healthcare_dataset.csv) is in the project folder.
3. Run the notebook to generate visuals and summaries.
4. View the EDA_Healthcare_Report.pdf for quick reference.

Summary

This project helps in identifying key patterns and potential risk factors by visualizing healthcare data. It provides insights into patient demographics, diagnosis trends, treatment types, and blood pressure variations.